

K.J. College Of Engineering And Management Research ,Pune.

Department of Engineering Sciences.

FACELOG

Project By :

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TRINITY

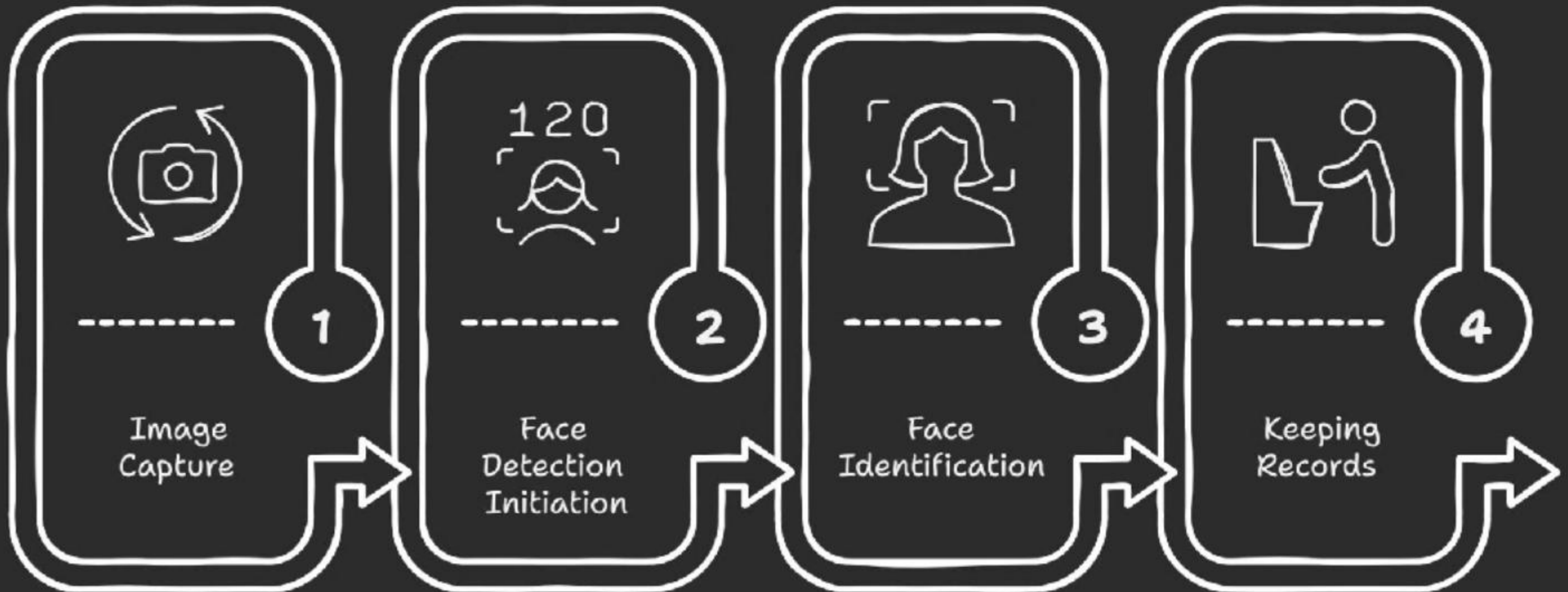
FaceLog

Facial Recognition-Based Student Attendance System.

Problem Statement :

Manual attendance marking systems are time-consuming, error-prone, and inefficient, especially for large groups. Issues like proxy attendance, data mismanagement, and human errors make tracking unreliable.

System Overview



Objectives:



Automation

- Eliminate manual recording of attendance.
- To develop an automated attendance system using face recognition.



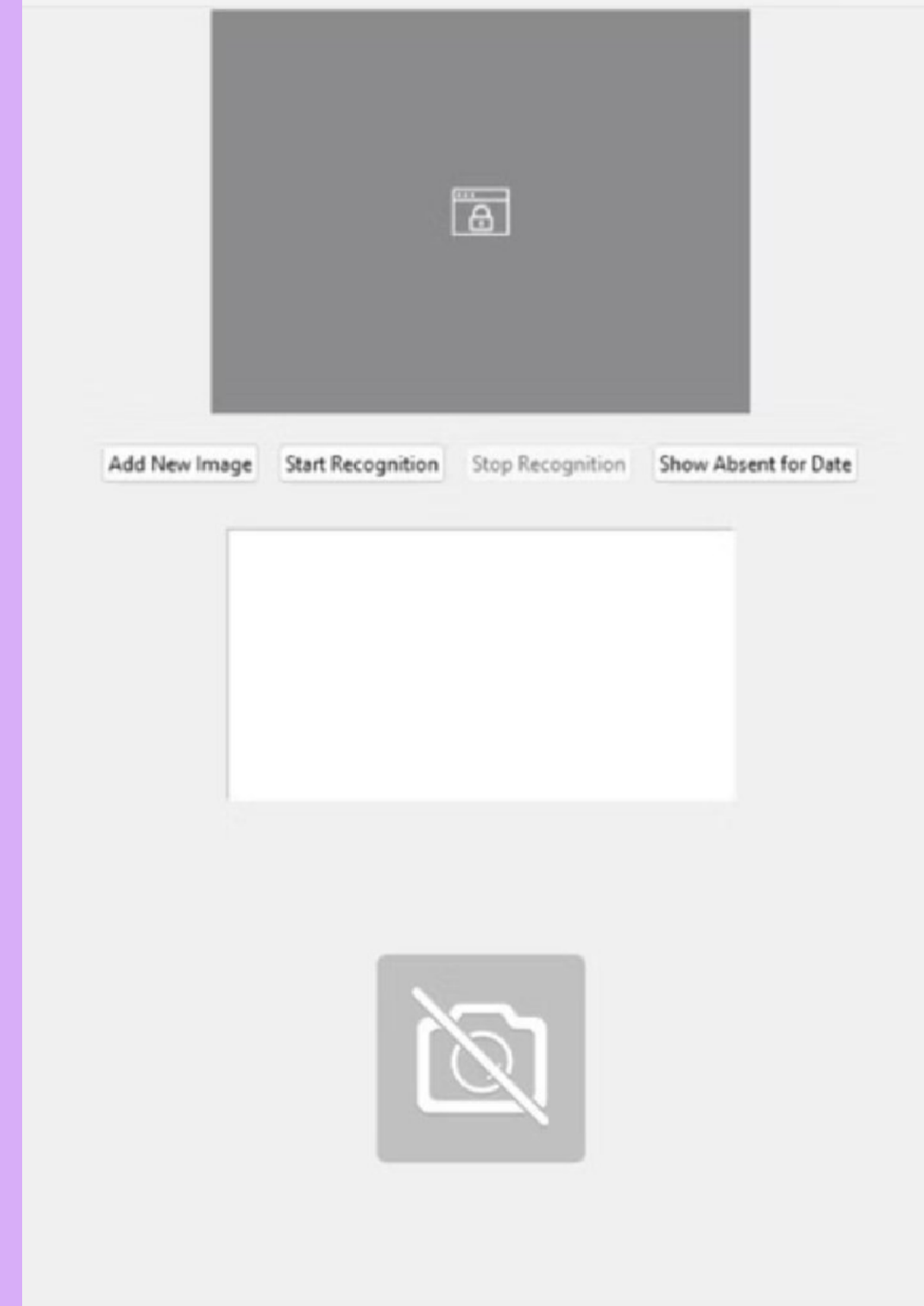
Accuracy

- Prevent proxy attendance, reduce human errors.
- To provide real-time attendance tracking and reporting



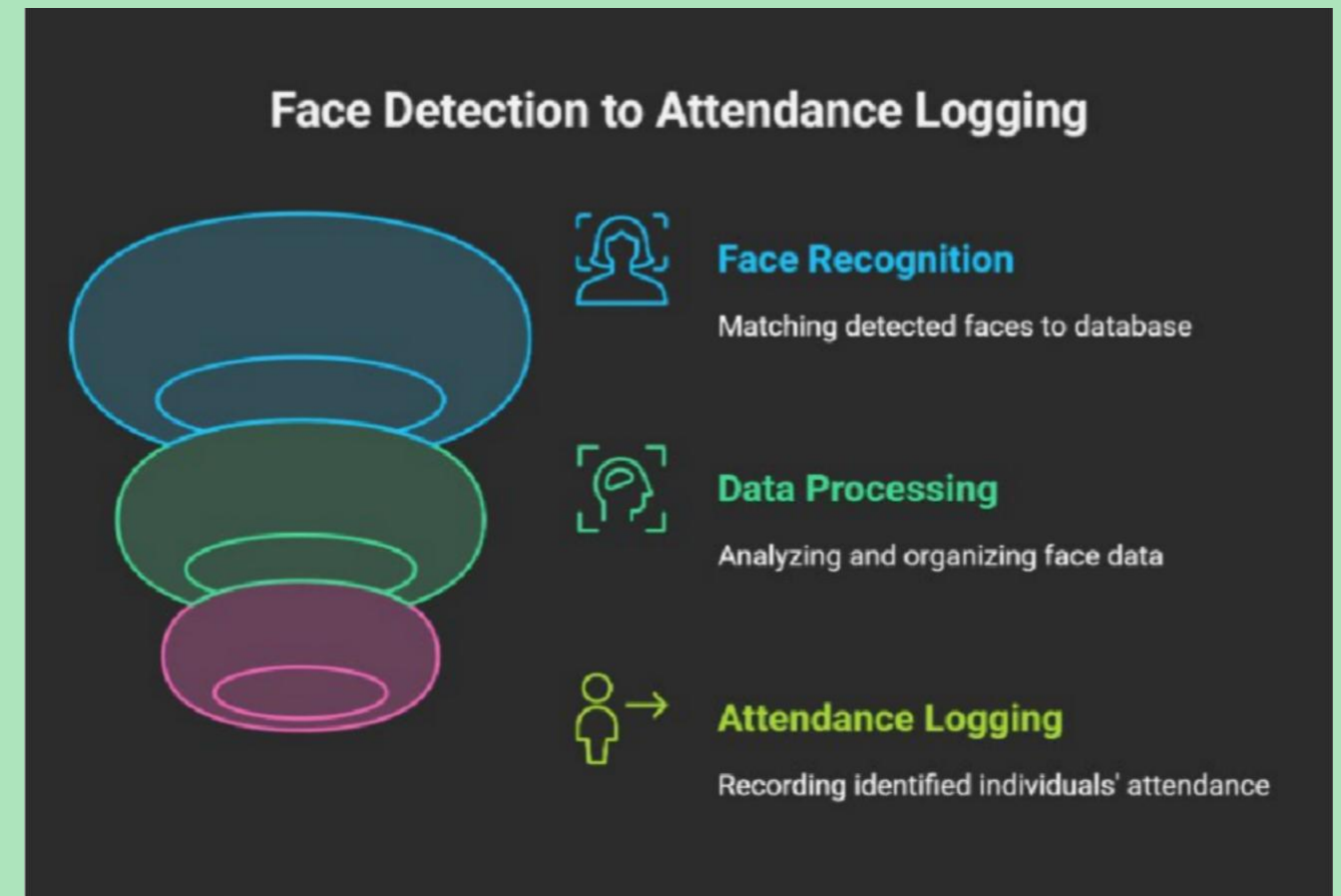
Efficiency

- Reduce time for attendance marking.
- Make it Easy to use and scalable.



System Capabilities

- **Capture images of students and store them in the database.**
- **Use face recognition to identify students in real-time.**
- **Export attendance data to CSV.**
- **Mark attendance and generate reports.**





The user interface displays attendance records. The server processes images and if not matched with Database No attendance is marked.

System Workflow

1

Image Acquisition

Camera captures student images (Input: Webcam feed./Enrollment Process for Students)

2

Face Detection

OpenCV detects faces.

3

User Interface

Tkinter used for GUI, Allows User to mark attendance by clicking 'Start Recognition' Button

4

Attendance Marking

System marks attendance record In Excell Sheet

5

Data Storage

Reports are Stored in a relational databases /Excell



Expected Outcomes

Accurate Records

Eliminates proxy attendance and errors.

Reduced Time

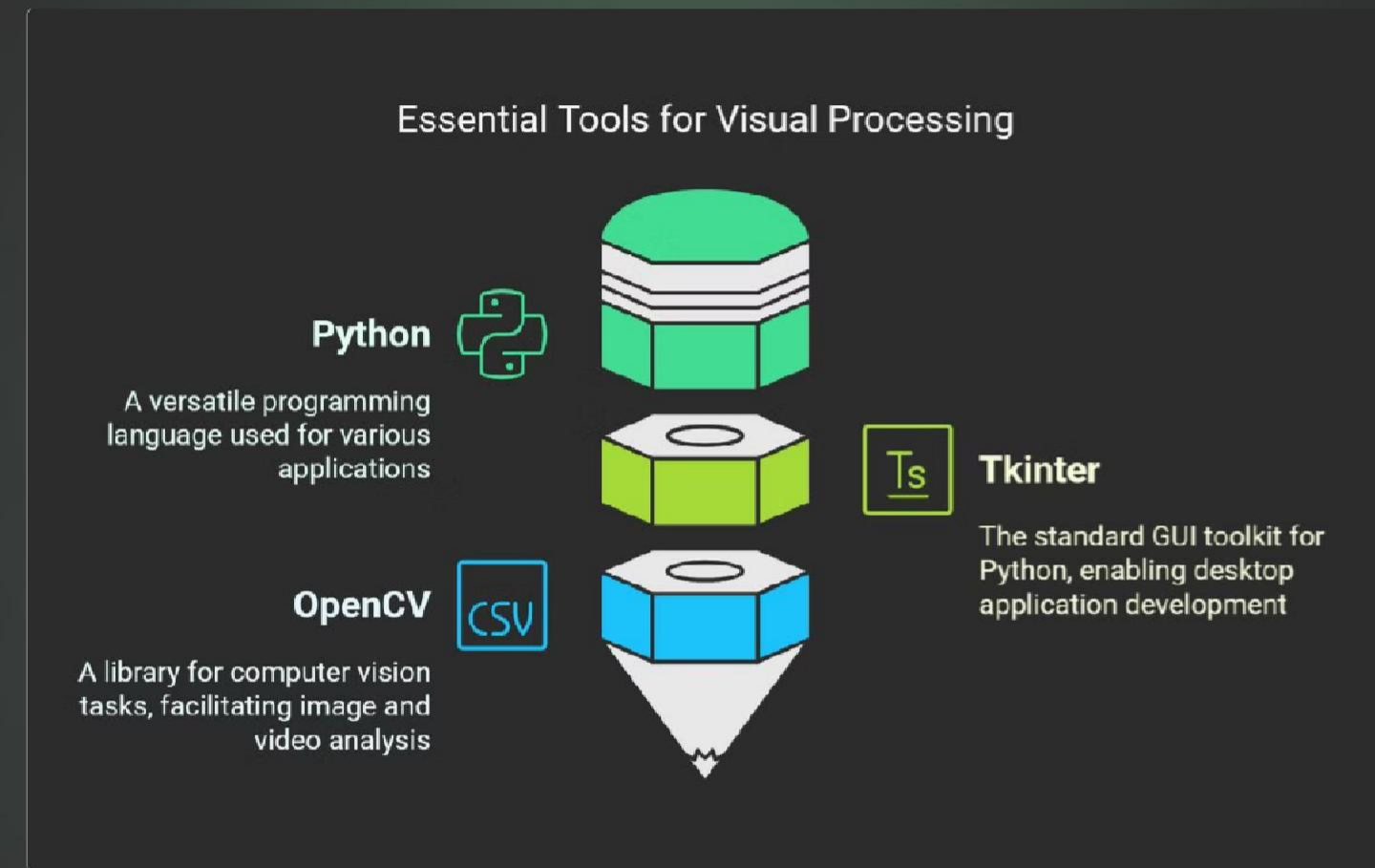
Significantly reduces attendance time.

Improved Efficiency

Streamlines process for students/instructors.

Components Used :

Programming Language	Python
Libraries/Frameworks	OpenCV, Face Recognition, Pandas, Tkinter
Tools	Visual Studio Code, Git
Hardware	Camera, Processing Unit



Challenges and Future Enhancements:

Challenges:

- Handling lighting and pose.
- Managing large datasets.
- Ensuring security/privacy.
- Managing lag in live video feed.

Future Enhancements:

- Integration with cloud storage for data backup
- Implementing emotion recognition.
- Developing a mobile app

Conclusion

Our system provides an efficient, accurate, automated solution.

It addresses limitations, saving time and reducing errors. Real-time monitoring and data analysis offers insights.

Future enhancements will improve its capabilities.

Components of a Professional Attendance Log



References

Python: <https://www.python.org/doc/>

OpenCV: <https://docs.opencv.org/>

Face Recognition Library: https://github.com/ageitgey/face_recognition

Tkinter (GUI Framework): <https://github.com/joeVenner/FaceRecognition-GUI-APP.git> **Pandas**

(Data Handling): <https://pandas.pydata.org/do>

